

SOFTWARE REQUIREMENTS SPECIFICATION

HackHub

Smart Hackathon & Project Management System

A professional frontend-first platform for discovering hackathons, showcasing projects, collaboration, and competition management.

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1.0.0	18/08/2026	Vikash Rai	Mr. Gautam Mukherjee	Initial complete SRS baseline

Table of Contents

1. Introduction
2. Overall Description
3. Stakeholders and User Roles
4. System Features and Functional Requirements
5. Non-Functional Requirements
6. External Interface Requirements
7. Data and Database Requirements
8. System Architecture and Technology Plan
9. Use Case Specification
10. Business Rules and Validation
11. UI/UX Requirements
12. Security and Privacy Requirements
13. Testing and Acceptance Requirements
14. Constraints, Assumptions and Dependencies
15. Future Scope
16. Traceability Matrix
17. Conclusion

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the complete functional, non-functional, interface, data, security, usability, testing, and future integration requirements for HackHub. It provides a common reference for the student development team, project guide, UI/UX designer, tester, and future backend/API developer.

1.2 Project Overview

HackHub is a modern frontend-based hackathon and project management platform. It is designed to help students and developers discover hackathons, understand problem statements and event details, manage participation, showcase projects, and interact with a developer-focused community. The system also includes an Admin module for managing hackathons, users, projects and platform content.

1.3 Objectives

- Centralize hackathon discovery and information in one platform.
- Provide a simple registration and login experience.
- Allow users to explore hackathons through search, filters and detailed pages.
- Allow participants to manage and showcase projects using GitHub repository links in the initial frontend scope.
- Provide an admin dashboard for platform management.
- Create a professional, responsive and maintainable UI suitable for academic demonstration.
- Keep the architecture ready for future database, API, authentication and notification integration.

1.4 Intended Audience

- GLA University faculty and project evaluators.
- Student developers and UI/UX designers.
- Software testers and project documentation teams.
- Future developers integrating APIs, databases and deployment.

1.5 Definitions

Term	Definition
Hackathon	A time-bound event where participants build solutions to given problems or themes.
Participant	Registered user who explores and participates in hackathons.
Project	A solution/application submitted or showcased by a participant/team.
Admin	Authorized person responsible for managing platform content and users.
SRS	Software Requirements Specification.
API	Application Programming Interface.
MVP	Minimum Viable Product.

2. Overall Description

2.1 Product Perspective

HackHub is planned as a frontend-first web application. During the current academic implementation, mock/static data may be used. The UI and modules are intentionally separated so that a future service/API layer and persistent database can be added without redesigning the complete application.

2.2 Product Functions

Module	Major Functions
Authentication	Login, signup, logout, validation, role-aware navigation
Home/Dashboard	Featured hackathons, quick actions, categories and platform highlights
Hackathon Discovery	Search, filter, sort and browse hackathons
Hackathon Details	Problem statement, rules, dates, eligibility, prizes, technologies and registration
Project Management	Create/manage project details, GitHub repository link, team information and submission status
Leaderboard	Display participant/team/project rankings and results where data is available
Profile	User information, projects, participation and account actions
Admin	Dashboard, hackathon CRUD, project moderation, user management and analytics
Community	Project showcase and developer-oriented discovery features

2.3 Scope

The MVP focuses on a realistic college-level platform. It does not require live payment processing, real-time judging, external hackathon APIs, advanced AI, or production-scale infrastructure.

2.4 Out of Scope for MVP

- Real payment gateway settlement.
- Automatic GitHub repository analysis.
- Real-time judge scoring and video judging.
- Live external event synchronization.
- Advanced AI recommendation engine.
- Real-time chat/video conferencing.

3. Stakeholders and User Roles

Stakeholder	Interest	Responsibilities / Access
Participant/User	Find and participate in hackathons	Register/login, browse events, register, manage profile, submit/showcase projects
Admin	Operate platform	Manage hackathons, projects, users, statuses and dashboard information
Project Guide	Academic supervision	Review requirements, progress and final implementation
Faculty/Evaluator	Assessment	Evaluate SRS, UI/UX, functionality,

		testing and presentation
Developer	Implementation	Build, test, maintain and integrate future APIs/database

3.1 Role-Based Access

Participant features must be accessible only to authenticated users where personal data or submissions are involved. Admin features must be protected by role-based authorization in the future backend; hiding an admin button in the frontend alone is not considered sufficient security.

4. System Features and Functional Requirements

4.1 Authentication

ID	Requirement	Priority
FR-01	System shall allow a new participant to create an account using required registration information.	High
FR-02	System shall validate email, password and mandatory fields.	High
FR-03	System shall allow registered users to log in.	High
FR-04	System shall show clear errors for invalid credentials or incomplete forms.	High
FR-05	System shall provide logout functionality.	High
FR-06	System shall support role-aware navigation for Participant and Admin.	High

4.2 Home and Dashboard

ID	Requirement	Priority
FR-07	System shall display featured/upcoming hackathons.	High
FR-08	System shall provide quick navigation to discovery and project areas.	Medium
FR-09	Participant dashboard shall show relevant registrations/projects when backend data is available.	Medium

4.3 Hackathon Discovery

ID	Requirement	Priority
FR-10	System shall display hackathons in reusable cards/list components.	High
FR-11	System shall support keyword search.	High
FR-12	System shall support filtering by category/technology, mode, duration/date or other available fields.	Medium
FR-13	System shall support sorting such as newest, deadline and popularity.	Medium
FR-14	System shall show hackathon status	High

	such as Upcoming, Open, Ongoing or Closed.	
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4.4 Hackathon Details and Registration

ID	Requirement	Priority
FR-15	System shall display title, organizer, dates, deadline, mode, eligibility, problem statement, rules, prizes and technologies where applicable.	High
FR-16	System shall provide a clear registration/participation CTA.	High
FR-17	System shall validate eligibility and required fields before registration in future API phase.	Medium
FR-18	System shall show registration confirmation/status to the participant.	High

4.5 Project Management

ID	Requirement	Priority
FR-19	Participant shall be able to create a project record or project draft.	High
FR-20	Project shall support title, description, technology stack, team information and project image where applicable.	High
FR-21	Participant shall be able to provide a GitHub repository link for the initial frontend implementation.	High
FR-22	System shall validate that required project fields are completed before submission.	High
FR-23	Participant shall be able to edit a draft project before final submission.	Medium
FR-24	System shall display project submission status.	Medium

4.6 Leaderboard and Showcase

ID	Requirement	Priority
FR-25	System shall display project/participant rankings when ranking data is available.	Medium
FR-26	System shall display showcased project cards with relevant information.	Medium
FR-27	System shall allow users to open project details from the showcase.	Medium

4.7 Profile

ID	Requirement	Priority
FR-28	Participant shall be able to view profile information.	High
FR-29	Participant shall be able to update	Medium

	permitted profile information.	
FR-30	Profile may display registered hackathons and submitted projects.	Medium

4.8 Admin Dashboard

ID	Requirement	Priority
FR-31	Admin shall see dashboard metrics such as users, hackathons and projects.	High
FR-32	Admin shall create hackathons.	High
FR-33	Admin shall edit hackathon details.	High
FR-34	Admin shall publish, close, deactivate or archive hackathons.	High
FR-35	Admin shall view and manage project submissions.	High
FR-36	Admin shall manage user records subject to authorization and privacy rules.	Medium
FR-37	Admin shall search/filter management tables.	Medium

4.9 Feedback

- Success, warning, error and information states shall be visually distinct.
- Destructive actions shall use confirmation dialogs where appropriate.
- Empty states shall explain what the user can do next.
- Future API failures shall show recovery-oriented messages rather than raw server errors.

5. Non-Functional Requirements

Category	Requirement
Usability	Core tasks should be understandable without training; navigation and CTAs must be consistent.
Performance	Avoid unnecessary assets and scripts; pages should load promptly under normal prototype conditions.
Responsiveness	Support desktop, tablet and mobile viewport sizes without horizontal overflow.
Accessibility	Use labels, readable contrast, keyboard focus states, semantic structure and alternative text.
Reliability	Invalid submissions must be blocked with actionable feedback.
Maintainability	Use modular HTML/CSS/JS and reusable patterns; avoid duplicated business logic.
Scalability	Architecture should support additional hackathons, users, projects and APIs.
Compatibility	Support current major browsers such as Chrome/Edge and a current Firefox/Safari equivalent where feasible.
Security	Authentication and authorization must be enforced securely in the backend phase.
Privacy	Collect and expose only information necessary for platform functions.

6. External Interface Requirements

6.1 User Interface

- Landing/Home page.
- Login and Signup pages.
- Hackathon listing and search/filter UI.
- Hackathon detail page.
- Registration flow.
- Participant dashboard/profile.
- Project creation/edit/submission UI.
- Project showcase and leaderboard.
- Admin dashboard and management tables.

6.2 Future API

Resource	Representative Operations
Auth	POST register, POST login, POST logout/refresh
Users	GET profile, PATCH profile, admin user listing
Hackathons	GET list/detail, POST create, PATCH update/status
Registrations	POST register, GET participant registrations
Projects	GET list/detail, POST create, PATCH update, POST submit
Leaderboard	GET rankings
Admin	GET dashboard metrics

6.3 Hardware

No specialized hardware is required. The system runs on standard laptops, desktops, tablets and smartphones with a modern browser.

6.4 Software

- HTML5, CSS3 and JavaScript for the current frontend implementation.
- Git/GitHub for version control and project repository links.
- Future backend may use Node.js/Express, Java/Spring Boot, Python or another approved stack.
- Future database may use MySQL, PostgreSQL, MongoDB or another approved technology.

7. Data and Database Requirements

The current prototype can use mock data. The logical database design below is intended for future API/database integration.

Entity	Important Fields	Purpose
User	user_id, name, email, password_hash, role, bio, skills, created_at, status	Accounts and role information
Hackathon	hackathon_id, title, organizer, description, start_date, end_date, deadline, mode, status, prizes	Hackathon/event information
ProblemStatement	problem_id, hackathon_id, title, description, category	Problems offered by an event

Registration	registration_id, user_id/team_id, hackathon_id, registered_at, status	Participation records
Project	project_id, hackathon_id, owner_id/team_id, title, description, tech_stack, github_url, status	Project/submission information
Team	team_id, name, leader_id, created_at	Team information
TeamMember	team_id, user_id, role	Team membership
Leaderboard	rank_id, hackathon_id, project_id, score, rank	Ranking/results
Notification	notification_id, user_id, message, type, read_status	Future notifications

7.1 Data Integrity

- Primary and foreign keys must preserve valid relationships.
- GitHub/project links must use an accepted URL format.
- A project must reference an existing hackathon and owner/team.
- Admin status changes must be auditable in the future backend.
- Private user data must not be exposed through public project pages.

8. System Architecture and Technology Plan

Layer	Responsibility	MVP	Future
Presentation	Pages, components, styles, responsive behavior	HTML/CSS/JS	Framework if required
Service	Data fetching and application services	Mock/local data	REST/JSON API
Business Logic	Validation, roles, status rules	Client-side basic validation	Backend service layer
Persistence	Users, hackathons, projects, registrations	Static/mock data	SQL/NoSQL database

8.1 Recommended Project Structure

index.html, signup.html, css/, JavaScript/js/, Images/, and future modules for dashboard, hackathons, projects and admin functionality. As the project grows, reusable components and service modules should be separated rather than placing all logic in one file.

8.2 Development Principles

- Separation of concerns.
- Reusable UI components/styles.
- Meaningful IDs/classes and file names.
- Git-based version control.
- No hard-coded secrets.
- API abstraction for future integration.

9. Use Case Specification

ID	Actor	Use Case	Precondition	Outcome
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UC-01	Participant	Register	No existing authenticated session	Account created or validation shown
UC-02	Participant/Admin	Login	Valid account	Correct role dashboard/navigation shown
UC-03	Participant	Discover Hackathons	Hackathon data available	Matching events displayed
UC-04	Participant	View Hackathon	Event exists	Details and CTA displayed
UC-05	Participant	Register for Hackathon	User authenticated; event open	Participation recorded
UC-06	Participant	Create Project	User has access to project workflow	Project draft created
UC-07	Participant	Submit Project	Required fields complete	Project submitted and status shown
UC-08	Participant	View Showcase	Projects available	Project details displayed
UC-09	Admin	Manage Hackathon	Admin authenticated	Hackathon created/updated/status changed
UC-10	Admin	Manage Projects	Admin authenticated	Submission reviewed/managed
UC-11	Admin	Manage Users	Admin authenticated	User records viewed/managed

9.1 Primary Use Case Flow – Project Submission

18. Participant logs in.
19. Participant opens My Projects.
20. Participant creates a new project.
21. Participant enters title, description and technologies.
22. Participant adds GitHub repository link and team information.
23. System validates required data.
24. Participant submits the project.
25. System displays submission status/confirmation.

10. Business Rules and Validation

- Only authenticated users may access private account and submission features.
- Only authorized admins may manage platform content.
- Closed hackathons must not accept new registrations.
- Required project fields must be completed before submission.
- A user may not create duplicate registrations for the same hackathon where duplicates are not permitted.
- A GitHub URL must be syntactically valid before acceptance.
- Project visibility may be controlled by submission/publication status.
- Admin operations that affect important records should be logged in future backend implementation.

11. UI/UX Requirements

11.1 Visual Language

- Modern, clean and engaging developer-focused design.
- Green as the documentation theme and a cohesive product accent.
- Strong visual hierarchy using bold headings, cards, badges and clear CTAs.
- Consistent spacing, typography, button states and form controls.

11.2 Navigation

- Public: Home, Explore Hackathons, Login, Signup.
- Participant: Dashboard, Hackathons, My Projects, My Registrations, Profile.
- Admin: Dashboard, Hackathons, Projects, Users and Settings/Reports where implemented.

11.3 Responsive Design

- Desktop: multi-column layouts and full navigation.
- Tablet: adaptive grids and compact navigation.
- Mobile: stacked content, collapsible navigation and full-width primary actions.

11.4 Feedback States

- Loading state.
- Empty state.
- Success state.
- Validation/error state.
- Confirmation modal for destructive actions.

12. Security and Privacy Requirements

- Passwords shall be hashed using a strong password-hashing algorithm in the backend phase.
- HTTPS shall be used for deployed systems.
- Role-based authorization shall be enforced server-side.
- Input validation and sanitization shall be applied to user-controlled data.
- API keys, database credentials and secrets shall be stored outside source code.
- Users shall only access their authorized private data.
- Admin functions shall not be protected solely by frontend route hiding.
- Future API should use secure authentication tokens/session management.

13. Testing and Acceptance Requirements

13.1 Testing Levels

- Unit testing for validation and utility functions.
- UI/component testing for forms and interactions.
- Integration testing for future API calls.
- System testing for complete user/admin workflows.
- Responsive and cross-browser testing.

- Security testing for authentication, authorization and input validation.

13.2 Acceptance Tests

Test ID	Scenario	Expected Result
AT-01	Valid signup	Account flow completes successfully
AT-02	Invalid login	Clear error shown; access denied
AT-03	Search hackathon	Relevant results displayed
AT-04	Apply filter	Only matching events displayed
AT-05	Open details	Complete event information visible
AT-06	Register for open hackathon	Registration confirmation/status shown
AT-07	Submit incomplete project	Validation blocks submission
AT-08	Submit valid GitHub link	Project is accepted as draft/submission
AT-09	Admin creates hackathon	New event appears in management/listing
AT-10	Non-admin opens admin route	Backend-enabled system denies access
AT-11	Mobile viewport	No major overflow; controls remain usable

14. Constraints, Assumptions and Dependencies

14.1 Constraints

- College project scope and development time.
- Limited team resources.
- Frontend-first implementation.
- No requirement for paid third-party services in MVP.

14.2 Assumptions

- Users have a modern web browser.
- Hackathon data is initially available through mock/static data.
- GitHub links are used for project repositories in the current no-backend scope.
- Technology choices can be adjusted with faculty approval.

14.3 Dependencies

- Development environment and browser.
- Git/GitHub repository.
- Future hosting/API/database services if deployed.

15. Future Scope

- Real backend and persistent database.
- Secure authentication and email verification.
- Team formation and teammate matching.
- AI-based hackathon and problem recommendations.
- Automatic GitHub repository insights.

- Online judging and scoring workflows.
- Real-time notifications and reminders.
- Community posts, comments and reactions.
- Certificates and participation records.
- Cloud deployment, analytics and CI/CD.

16. Traceability Matrix

Objective	Requirements	Validation
Easy hackathon discovery	FR-10 to FR-14	Search/filter/status tests
Clear event understanding	FR-15 to FR-18	Details and registration tests
Project showcase/submission	FR-19 to FR-27	Project workflow tests
Account management	FR-01 to FR-06, FR-28 to FR-30	Authentication/profile tests
Admin management	FR-31 to FR-37	Admin workflow and authorization tests
Professional UX	NFR + UI/UX requirements	Responsive/accessibility/usability tests
Future scalability	Architecture + database requirements	API/database integration review

17. Conclusion

HackHub provides a realistic and professional foundation for a college-level hackathon and project management platform. This SRS defines the current frontend MVP while deliberately preparing the system for future API, database, authentication, notifications and advanced community capabilities. The requirements are intended to guide implementation, UI/UX design, testing, faculty evaluation and final project presentation.

Approval and Review

Role	Name	Signature	Date
Prepared By	Vikash Rai		18/08/2026
Project Guide	Mr. Gautam Mukherjee		
HOD	Mr. Kumar Sunil		

— End of Software Requirements Specification —